

EXP NO 15: Implement Iris Flower Classification using Naive Bayes classifier

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AIM

To implement Iris Flower Classification using the Naive Bayes classifier and classify flowers into Setosa, Versicolor, and Virginica based on their features.

PROCEDURE

1. Load the Iris dataset containing sepal and petal measurements.
2. Preprocess the dataset and divide it into training and testing data.
3. Create a Gaussian Naive Bayes classifier and train it using the training data.
4. Predict the flower species using the trained model.
5. Evaluate the classifier using accuracy and confusion matrix.

CODE:

```
from sklearn.datasets import load_iris

from sklearn.model_selection import train_test_split

from sklearn.naive_bayes import GaussianNB

from sklearn.metrics import accuracy_score, confusion_matrix


# Load Iris dataset

iris = load_iris()

X = iris.data

y = iris.target


# Split dataset

X_train, X_test, y_train, y_test = train_test_split(

    X, y, test_size=0.2, random_state=42

)
```

```

# Create and train Naive Bayes model

model = GaussianNB()

model.fit(X_train, y_train)


# Predict

y_pred = model.predict(X_test)


# Display results

print("Predicted Classes:", y_pred)

print("Actual Classes: ", y_test)

print("Accuracy:", accuracy_score(y_test, y_pred))


print("\nConfusion Matrix:")

print(confusion_matrix(y_test, y_pred))

```

OUTPUT:

```

Predicted Classes: [1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0
Actual Classes:    [1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0
Accuracy: 1.0

Confusion Matrix:
[[10  0  0]
 [ 0  9  0]
 [ 0  0 11]]

```

RESULT:

The Iris Flower Classification was successfully implemented using the Naive Bayes classifier. The model classified the three iris species—Setosa, Versicolor, and Virginica—with an accuracy of 100% on the test data.

